Learning to Segment Referred Objects from Narrated Egocentric Videos

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Background: Egocentric Vision + Vision-Language are Transforming Perception

- We leverage first-person cameras that capture rich, hands-on human–object interactions.
- We train on massive narrated video corpora (e.g., Ego4D) to fuel multimodal learning.
- We build on vision–language models to enable open-world, instruction-following perception.

We define Narration-based Video Object Segmentation (NVOS).

- We take as input a short egocentric clip and its activity narration.
- We aim to segment each narrated object instance per frame.
- We train without spatial annotations (weak supervision).

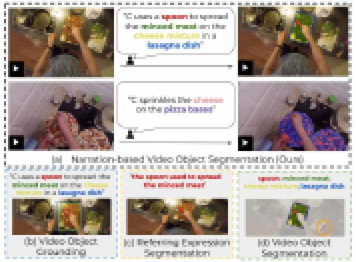

Figure: NVOS task: segment narrated objects per frame.

Why Object-Level Understanding Matters in Egocentric Video

- We need instance-level disambiguation, not just class labels.
- We handle occlusions, overlaps, and non-axis-aligned shapes.
- We produce pixel-accurate masks for tracking, manipulation, and safety.

Our Core Idea: We learn weakly supervised pixel-level grounding with ROSA

- We leverage CLIP [36] to align noun phrases and region masks.
- We train with video–narration contrastive objectives only.
- We infer region–phrase links using latent alignments.

Contributions at a Glance

- We define NVOS: phrase-to-pixel alignment in egocentric videos.
- We propose ROSA: mask-guided CLIP [36] with global–local contrastive learning.
- We introduce temporal adaptive pooling for robust multi-frame evidence.
- We curate VISOR-NVOS with object-linked narrations.
- We show state-of-the-art zero-shot pixel-level grounding.

How We Train ROSA: Pipeline Overview

- We generate class-agnostic mask proposals with SAM [21].
- We encode regions and phrases with context-aware CLIP [36].
- We compute similarities and aggregate them over time.
- We train with global and local contrastive losses.

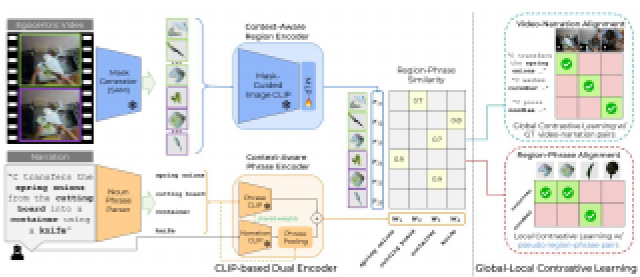

Figure: ROSA framework: region–phrase embeddings and global–local contrastive learning.

Mask Proposals: High-Recall, Class-Agnostic Regions

- We use SAM [21] to propose per-frame instance masks.
- We filter proposals with confidence and NMS to keep high recall without labels.
- We preserve weak supervision while providing a strong upper bound.

Context-Aware Region Encoder: Mask-Guided CLIP

- We run CLIP [36] ViT up to the penultimate layer on the full frame.
- We restrict final-layer CLS attention to the proposal region via the mask.
- We project features through an MLP to compact region embeddings.

Context-Aware Phrase Encoder

- We extract noun phrases from narration (e.g., spaCy).
- We compute a localized phrase embedding (phrase alone).
- We compute a narration-aware embedding (contextualized).
- We average both to obtain a robust phrase representation.

Region–Phrase Similarity and Frame Affinity

- We compute cosine similarity between region and phrase embeddings.
- We weight similarities by SAM mask confidence.
- We select the best region per frame for each phrase.

Temporal Adaptive Pooling

- We perform softmax-weighted pooling over frame affinities.
- We increase temperature during training to broaden evidence.
- We start by focusing on the best frame, then integrate many.

Global–Local Contrastive Learning: Video–Narration Alignment

- We aggregate frame–phrase scores to video–phrase scores.
- We average over phrases to obtain a video–narration score.
- We apply symmetric InfoNCE across batch positives/negatives.

Global–Local Contrastive Learning: Region–Phrase Alignment

- We create pseudo labels via the best region per frame.
- We apply region-to-phrase InfoNCE with hard negatives.
- We use phrase-to-region MIL-NCE across frames.
- We sample negatives from aligned regions of other phrases.

Inference: Zero-Shot Pixel-Level Grounding

- We choose the max-similarity proposal per frame and phrase.
- We output its mask with no temporal post-processing.
- We generalize zero-shot to new datasets and views.

Dataset & Benchmark: VISOR-NVOS

- We pair VISOR masks with 14.6k object-centric narrations.
- We provide explicit links from phrases to object IDs.
- We support fine-grained, multi-instance evaluation.

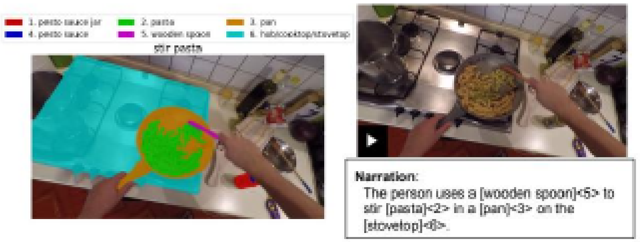

Figure: VISOR-NVOS example: narrations link phrases to object IDs.

Experimental Setup

- We train on 250k Ego4D narrated clips with 4 FPS sampling.
- We supervise only with video–narration pairs; no spatial labels.
- We evaluate on VISOR-NVOS and VOST (zero-shot).
- We report J, F, J&F; Junion and Jins for VOST.

Main Results: State-of-the-Art Zero-Shot Grounding

- We outperform weakly and strongly supervised baselines on VISOR-NVOS.
- We show competitive zero-shot performance on VOST.
- We demonstrate strong generalization across domains and views.

Table: Table 1. Comparison of grounding performance between our framework and strong baselines on our NVOS benchmarks.

	Supervision				VISOR-NVOS			VOST
Method	Cross-Modal	Mask Annotation	Ego4D†	J	F	J &F	Junion	Jins
SAM [21] upper bound				70.3	75.6	73.0	62.2	65.8
	Trained w/ labeled regions							
ODISE [52]	mask-text	class-specific		29.0	32.8	30.9	17.6	21.1
GroundedSAM [21, 27]	bbox-text class-agnostic			37.3	41.8	39.5	21.8	25.3
		Trained w/o labeled regions						
SAM [21] + CLIP [36] (ViT-B/16)	image-text	class-agnostic		22.2	25.8	24.0	16.5	19.2
CoMMa† [43] + SAM [21]	video-narration	✓ class-agnostic		15.3	25.3	20.3	9.4	11.5
ROSA (ViT-B/16)	video-narrationclass-agnostic		✓	34.9	41.2	38.1	22.2	25.4
ROSA (ViT-L/14)	video-narration	✓ class-agnostic		38.7	46.0	42.4	23.2	26.7

Baselines: SAM [21], CLIP [36], ODISE [52], GroundedSAM [27], CoMMa [43].

Qualitative Results: Instance Disambiguation and Robustness

- We disambiguate multiple same-class instances via phrases.
- We handle large appearance and shape changes.
- We produce fine-grained, pixel-accurate masks.

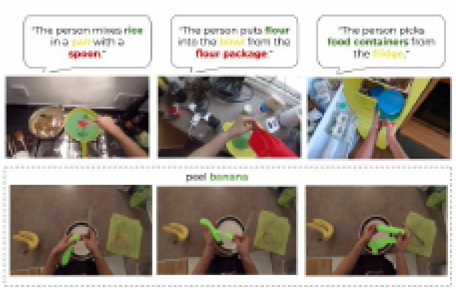

Figure: Qualitative results on VISOR-NVOS and VOST.

Failure Modes and Headroom

- We observe partial masks and size biases on small objects.
- We face plural phrases and part–whole ambiguities.
- We do not explicitly enforce temporal consistency.
- The SAM [21] upper bound highlights remaining potential.

Open Questions

- How can we enforce temporal consistency during training and inference?
- How do we robustly disambiguate plurals, parts, and attributes?
- How well can we scale beyond cooking to broader domains?
- What interfaces make these models safe and useful in assistive/embodied settings?

Questions & Discussion

- Thank you for your attention!
- Questions and feedback are welcome.

Acknowledgments

- Thanks to collaborators, dataset creators, and funding sources.
- Community resources: Ego4D, VISOR, and open-source toolkits.

Backup: Cross-Domain Generalization (YouCook2-BB)

- We transfer zero-shot from egocentric to third-person videos.
- We observe substantial gains in box and point accuracy.
- We demonstrate versatile region–phrase alignment.

Table: Table 3. Object grounding evaluation on YouCook2-BB.

		box accuracy			
Method	macro	micro			
Trained on YouCook2					
Zhou et al. [64]	35.08	42.42	Method	point acc.	
NAFAE [41]	40.71	46.33	CoMMa†	53.56	
STVG [56]	41.67	48.22	CoMMa [43]	59.25	
SCL [48]	42.80	48.60	Ours	69.35	
Zero-Shot		(b) point prediction evaluation			
CoMMa†+SAM	6.63	8.98			
Ours	37.93	44.96			

Baselines: Zhou et al. [64], NAFAE [41], STVG [56], SCL [48], CoMMa [43].

Backup: Ablations – What Drives Performance?

- We quantify the impact of global VNA, local R2P/P2R, negative sampling, and temporal pooling.

Table: Table 2. Ablation studies on the Global–Local Contrastive Learning framework on the VISOR-NVOS test split.

VNA	R2P	P2R	J	F	J &F	Negative samples	J	F	J &F		J	F	J &F
✓			24.833.4	31.3 39.6	28.136.0	same	27.5	35.0	31.3	random frame	33.3	39.8	36.6
✓	✓		33.8	40.1	37.0	all	27.4	34.8	31.1	max pooling	33.7	40.5	37.1
✓		✓	33.8	40.6	37.3	other	26.5	34.0	30.3	softmax	33.6	40.4	37.0
✓	✓	✓	34.9	41.2	38.1	aligned (ours)	34.9	41.2	38.1	adaptive (ours)	34.9	41.2	38.1
	(a) Impact of various losses.			(b) Ablation for negative region sampling.				(c) Ablation of temporal aggregation functions.					